

# CS & IT ENGINEERING

DISCRETE MATHS  
SET THEORY



**Lecture No. 8**



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TOPICS TO BE  
COVERED

01 Basics of relations

02 Types of relations

03 Number Of relations

# Relations



Reflexive:  $\rightarrow \forall a \in A \ aRa$

Irreflexive:  $\forall a \ (a, a) \notin R$

Symmetric:  $\forall a \forall b \ ((a, b) \in R \rightarrow (b, a) \in R)$

Anti:  $((a, b) \in R \wedge (b, a) \in R \rightarrow a = b)$

Asy:  $((a, b) \in R \rightarrow (b, a) \notin R)$



# Relations

$A \rightarrow$  non empty set

$A = \{1, 2, 3\}$

$R_1 \rightarrow$   
reflexive



$R_2$   
 $\downarrow$   
reflexive

$R_1 \cup R_2$

$$\left\{ \boxed{11 \ 22 \ 33}, ( ) ( ) \right\} \cup \left\{ \boxed{11 \ 22 \ 33} ( ), ( ) \right\}$$

$$= \left\{ \boxed{11 \ 22 \ 33} ( ) ( ) ( ) \dots \right\}$$

# Relations

|                          | Union. | Intersection. |
|--------------------------|--------|---------------|
| $R_1, R_2$<br>reflexive. | ✓      | ✓             |
| $R_1, R_2$<br>symmetric. | ✓      | ✓             |
| $R_1, R_2$<br>Anti°      | ✗      | ✓             |
| $R_1, R_2$<br>Asymmetric | ✗      | ✓             |
| $R_1, R_2$ Irreflexive   | ✓      | ✓             |
| $R_1, R_2$ Transitive    | ✗      | ✓             |



# Relations



$A \rightarrow$  non empty set.

$R_1 \rightarrow$  symme'

$R_2 \rightarrow$  symmetric.

$$\{ \underset{\text{a}}{(13)} \underset{\text{a}}{(31)} \} \cap \{ \underset{\text{b}}{(23)} \underset{\text{b}}{(32)} \}$$

$$R_1 \cap R_2 = \{ \mid \checkmark \}$$

$$R_1 \cap R_2 = \{ (11) \mid \checkmark \}$$

# Relations



$R_1 \rightarrow \text{Anti}$

$$R_1 = \{ (13) \} \checkmark$$

$R_2 \rightarrow \text{Anti}$

$$R_2 = \{ (31) \} \checkmark$$

Anti

$\rightarrow$  no flipping  
 $\rightarrow$  allows same element

$$R_1 \cup R_2 = \{ (13) (31) \} \times$$

$$R_1 \cap R_2 = \emptyset \checkmark$$



# Relations

Transitive Relation :

$$\forall a \forall b \forall c \{ (a, b) \in R \wedge (b, c) \in R \longrightarrow (a, c) \in R. \}$$

OR.

$$\forall a \forall b \forall c \{ a R b \wedge b R c \longrightarrow a R c \}$$



# Relations



$$R_1 = \{ \} \checkmark$$

$$(a,b) \in R \wedge (b,c) \in R \rightarrow (a,c) \in R.$$

$$\begin{array}{c} F \wedge F \rightarrow \\ \hline F \rightarrow \\ \hline \text{True.} \end{array}$$

$$\begin{array}{l} a=1 \\ b=3 \end{array}$$

$$R_2 = \{ (1,3) \} \checkmark$$

$$(a,b) \in R \wedge (b,c) \in R \rightarrow (a,c) \in R.$$

$$(1,3) \in R \wedge F \rightarrow$$

$$F \rightarrow$$

True

# Relations

$$R_2 = \{ (11) \} \checkmark$$

$$(a, b) \in R \wedge (b, c) \in R \rightarrow (a, c) \in R.$$

$$(1, 1) \in R \wedge (1, )$$

$$a = 1$$

$$b = 1$$

$$R_3 = \{ (13) (31) (11) \}$$



# Relations



$$R_3 = \{ (1,3) (3,1) (\underline{1,1}) \} \rightarrow R_3 = \{ (\underline{3,1}) (\underline{1,3}) (\cancel{1,1}) \}$$

$$(a,b) \in R \wedge (b,c) \in R \rightarrow (a,c) \in R.$$

$$(3,1) \wedge (1,3) \rightarrow (3,3) \in R$$

$$(1,3) \in R \wedge (3,1) \in R \rightarrow (1,1) \in R.$$

$$a=1 \quad c=1$$

$$b=3$$

$$a=1$$

$$b=3$$

$$c=1.$$

$$R_3 = \{ (1,3) (3,1) (1,1) (3,3) \}$$



# Relations

$$A = \{1, 2, 3, 4\}$$

$$\begin{aligned} 1 &\rightarrow \{(1, 1)\} \\ 2 &\rightarrow \{(2, 2) (2, 1) (2, 3) (2, 4)\} \\ 3 &\rightarrow \{(3, 3) (3, 2) (3, 1) (3, 4)\} \end{aligned}$$

$$R = \{(1, 1) (2, 1) (2, 2) (2, 3) (2, 4) (3, 1) (3, 2) (3, 3) (3, 4)\}$$





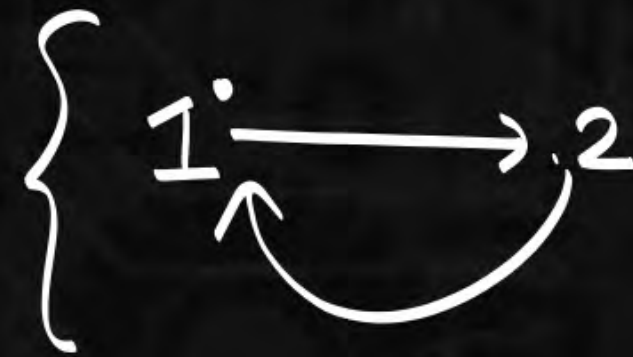
# Relations



$$R = \{ (1,2), (2,1) \}$$

Check it is transitive  
or not

$$R = R'$$



$$1 \rightarrow (1,2), (1,1)$$

$$2 \rightarrow (2,1), (2,2)$$

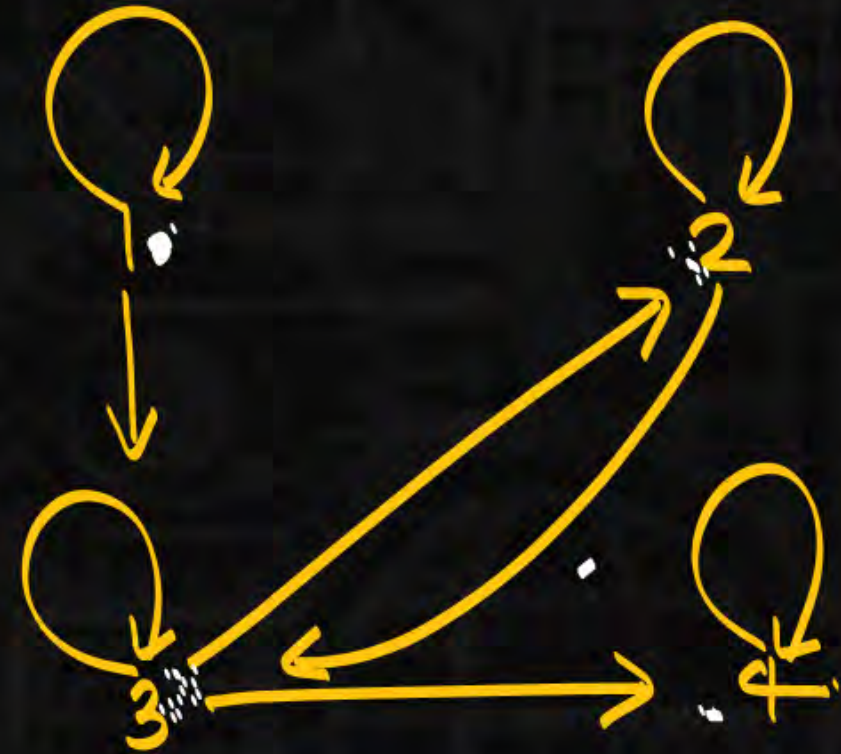
$$R' = \{ (1,2), (1,1), (2,1), (2,2) \}$$

# Relations



$$R = \left\{ \begin{array}{l} \underline{(1,1)} \quad (2,2) \quad (3,3) \quad (4,4) \\ (1,3) \quad (3,2) \quad (3,4) \\ (2,3) \end{array} \right\}$$

$$R \neq R^{exp}$$



$$\begin{aligned} 1 &\rightarrow (1,1) \quad (1,3) \quad \underline{(1,2)} \quad (1,4) \rightarrow 4 \\ 2 &\rightarrow (2,2) \quad (2,3) \quad (2,4) \checkmark \rightarrow 3 \\ 3 &\rightarrow (3,3) \quad (3,2) \quad (3,4) - 3 \\ 4 &\rightarrow (4,4) - 1 \end{aligned}$$



# Relations



$$R_1 = \{ (a, b) \mid a + b \leq 3 \}$$

$$(a, b) \in R \wedge (b, c) \in R \rightarrow (a, c) \in R.$$

$$a + b \leq 3 \wedge b + c \leq 3 \rightarrow a + c \leq 3$$

$$2 + 0 \leq 3 \wedge 0 + 2 \leq 3$$

$$a = 2, b = 0$$

$$(2, 0) \in R$$

$$b = 0, c = 2$$

$$(0, 2) \in R \rightarrow 2 + 2 \leq 3.$$

$$\text{Result: } \left\{ \overset{\text{present}}{\overbrace{(2, 0)}} \overset{\text{present}}{(0, 2)} \right\}$$

expect (2, 2)  
will be absent.

(not transitive)

# Relations



$$R_2 = \{ (a, b) \mid \underline{gcd(a, b) = 1} \}$$

$$gcd(2, 4) = 2.$$

absent

$$(a, b) \in R \wedge (b, c) \in R \rightarrow (a, c) \in R.$$

$$gcd(2, 3) = 1.$$



$$(2, 3) \in R \wedge (3, 8) \in R$$

$$gcd(a, b) = 1 \wedge gcd(b, c) = 1 \rightarrow gcd(a, c) = 1.$$

$$\underline{gcd(2, 3) = 1} \wedge \underline{gcd(3, 8) = 1} \rightarrow gcd(2, 8) = 2 \text{ (absent)} \quad \underline{\text{not transitive}}$$



# Relations



$R_1 \rightarrow \text{Transitive.}$

$$\underline{R_1 = \{(13)\} \checkmark} \cup$$

$R_2 \rightarrow \text{Transitive.}$

$$\underline{R_2 = \{(31)\} \checkmark}$$

$R_1 \cup R_2 \rightarrow \underline{\text{Transitive.}}$

$$\underline{\{(13)(31)\} \times}$$

